WORKED SOLUTION

STEP II 2025, Question 12

Probability and Statistics • Poisson Distribution

STEP II 2025, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

For a Poisson distribution, consecutive probabilities have the simple ratio $p_r/p_{r-1} = \lambda/r$, so every difference d_r can be written as a known probability times a simple rational factor in r. The sign of d_r tells us where p_r rises and falls; going one level further, the sign of the *second* difference $d_{r+1} - d_r$ tells us where d_r rises and falls, and that sign is controlled by a quadratic in r. The awkward-looking number $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}}$ is just (one more than) a root of that quadratic.

Solution

Throughout, $p_r = \mathrm{e}^{-\lambda}\lambda^r/r!$, so for $r \geq 1$

$$p_r = \frac{\lambda}{r}\,p_{r-1}, \qquad p_{r-1} = \frac{r}{\lambda}\,p_r,$$

and every p_r is strictly positive.

Part (i)

For $r \geq 1$,

$$d_r = p_r - p_{r-1} = p_{r-1}\left(\frac{\lambda}{r} - 1\right) = \frac{p_{r-1}}{r}\,(\lambda - r).$$

Since $p_{r-1}/r > 0$, the sign of d_r is the sign of $\lambda - r$, and d_r is never zero because λ is not an integer. Let $m = \lfloor \lambda \rfloor$, the largest integer below λ, so that $\lambda - 1 < m < \lambda$ (both inequalities strict because $\lambda \notin \mathbb{Z}$). Then

- for $1 \leq r \leq m$ we have $r < \lambda$, so $d_r > 0$ and $p_0 < p_1 < \cdots < p_m$;
- for $r \geq m+1$ we have $r > \lambda$, so $d_r < 0$ and $p_m > p_{m+1} > p_{m+2} > \cdots$.

(If $\lambda < 1$ then $m = 0$ and the first list is empty.) Hence $p_r < p_m$ for every $r \neq m$: m is the unique most likely value, and $\lambda - 1 < m < \lambda$.

Part (ii)

Now look at how d_r itself changes. For $r \geq 1$, put $e_r = d_{r+1} - d_r = p_{r+1} - 2p_r + p_{r-1}$. Writing everything in terms of p_r,

$$e_r = p_r\left(\frac{\lambda}{r+1} - 2 + \frac{r}{\lambda}\right) = \frac{p_r}{\lambda(r+1)}\Big(\lambda^2 - 2\lambda(r+1) + r(r+1)\Big).$$

The prefactor is positive, so the sign of e_r is the sign of

$$Q(r) = r^2 + (1-2\lambda)r + \lambda^2 - 2\lambda.$$

This quadratic has discriminant $(1-2\lambda)^2 - 4(\lambda^2 - 2\lambda) = 1 + 4\lambda$, so its roots are

$$\alpha = \lambda - \tfrac{1}{2} - \sqrt{\lambda + \tfrac{1}{4}}, \qquad \beta = \lambda - \tfrac{1}{2} + \sqrt{\lambda + \tfrac{1}{4}},$$

and $Q(r) < 0$ exactly when $\alpha < r < \beta$. Thus

$$d_{r+1} < d_r \iff \alpha < r < \beta, \qquad d_{r+1} > d_r \iff r < \alpha \text{ or } r > \beta.$$

Note that $\beta + 1 = \lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}}$, which is not an integer, so β is not an integer, and $k = \lfloor \beta + 1 \rfloor = \lfloor \beta \rfloor + 1$ is exactly the integer with

$$k < \lambda + \tfrac{1}{2} + \sqrt{\lambda + \tfrac{1}{4}} < k + 1.$$

Ask yourself Where should we look for the minimum of d_r at all, given that d_r also increases when $r < \alpha$?

From part (i), $d_r > 0$ for $r < \lambda$ and $d_r < 0$ for $r > \lambda$, so the minimum must be among the negative terms, $r \geq m+1$. Also $\beta > \lambda$ (since $\sqrt{\lambda + \frac{1}{4}} > \frac{1}{2}$) and $\alpha < \lambda$, so every $r \geq m+1$ satisfies $r > \alpha$, and $k > \beta > \lambda$ gives $k \geq m+1$. On the integers $r \geq m+1$ we therefore have:

- for $m+1 \leq r \leq k-1$: $r < \beta$, so $e_r < 0$, giving $d_{m+1} > d_{m+2} > \cdots > d_k$;
- for $r \geq k$: $r > \beta$, so $e_r > 0$, giving $d_k < d_{k+1} < d_{k+2} < \cdots$.

(The first list is empty when $k = m+1$.) So d_k is strictly smaller than every other negative d_r, and smaller than all the positive ones too. The minimum of d_r occurs at $r = k$, as required. In words: p_r falls most steeply between $r = k-1$ and $r = k$.

Part (iii)

First, if $\lambda < 1$ then every d_r is negative by part (i). But $d_r \to 0$ as $r \to \infty$ (because $p_r \to 0$), so for any r there is a later term larger than d_r: no maximum is attained at all, and certainly not at $r = 1$. Hence we need $\lambda > 1$ ($\lambda = 1$ is excluded since λ is not an integer).

Now take $\lambda > 1$, so $d_1 > 0$. The positive terms are $d_1, \ldots, d_m$, and any maximum lies among them.

- If the maximum is at $r = 1$, then $d_2 < d_1$, i.e. $e_1 < 0$.
- Conversely, if $e_1 < 0$ then $\alpha < 1 < \beta$, so every integer r with $1 \leq r < \beta$ has $e_r < 0$. Since $m < \lambda < \beta$, this gives $d_1 > d_2 > \cdots > d_m$, and all later terms are negative. So the maximum is at $r = 1$.

So, for $\lambda > 1$, the maximum is at $r = 1$ exactly when $e_1 < 0$, i.e. when $Q(1) < 0$:

$$Q(1) = 1 + 1 - 2\lambda + \lambda^2 - 2\lambda = \lambda^2 - 4\lambda + 2 < 0 \iff 2 - \sqrt{2} < \lambda < 2 + \sqrt{2}.$$

Combining with $\lambda > 1$ (and $2 - \sqrt{2} < 1$), the condition is

$$1 < \lambda < 2 + \sqrt{2}.$$

Part (iv)

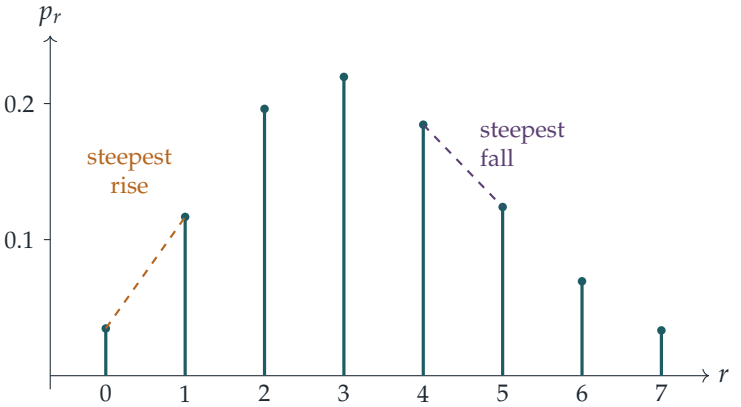
With $\lambda = 3.36$ we have $\lambda + \frac{1}{4} = 3.61 = 1.9^2$, so:

- the mode is $m = 3$, by part (i);
- $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} = 3.86 + 1.9 = 5.76$, so $k = 5$: the steepest fall is from p_4 to p_5;
- $1 < 3.36 < 2 + \sqrt{2} \approx 3.41$, so the steepest rise is from p_0 to p_1 (only just: $\alpha = 0.96$ is close to 1, so d_1 and d_2 are nearly equal).

Using $p_r = \frac{3.36}{r} p_{r-1}$ and $p_0 = \mathrm{e}^{-3.36} \approx 0.035$ gives the approximate values

$$p_0, \ldots, p_7 \approx 0.035,\ 0.117,\ 0.196,\ 0.220,\ 0.185,\ 0.124,\ 0.069,\ 0.033.$$

The sketch should show a rise that is steep from the start and then flattens to a peak at $r = 3$, followed by a fall that gets steeper, is steepest between $r = 4$ and $r = 5$, and then eases off.



Checks

- Computing p_r and d_r for 20,000 random values of $\lambda \in (0, 40)$ confirms that the mode is $\lfloor \lambda \rfloor$, that the minimum of d_r is at $\lfloor \lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} \rfloor$, and that (for $\lambda > 1$)

the maximum of d_r is at $r = 1$ exactly when $\lambda < 2 + \sqrt{2} \approx 3.414$.

- For $\lambda = 3.36$: $d_1 \approx 0.0820$ and $d_2 \approx 0.0794$, so d_1 is (narrowly) the largest; $d_4, d_5, d_6 \approx -0.035, -0.061, -0.055$, so d_5 is the smallest, as predicted.
- Small λ: as $\lambda \to 0$, $\lambda + \frac{1}{2} + \sqrt{\lambda + \frac{1}{4}} \to 1$, so $k = 1$. Indeed then nearly all the probability is at 0, and the biggest drop is the first one, $p_1 - p_0$.

Common slips

- Forgetting that d_r also *increases* for $r < \alpha$, and so not explaining why that region cannot contain the minimum (it is where $d_r > 0$).
- In part (iii), checking only $d_2 < d_1$ and missing the case $\lambda < 1$, where every d_r is negative and the sequence never attains a maximum.
- Writing $\lambda - 1 \leq m \leq \lambda$: the inequalities are strict precisely because λ is not an integer.

Beyond STEP

The differences d_r and e_r are discrete analogues of the first and second derivatives, and k marks a discrete "point of inflection" of the Poisson distribution. For large λ the Poisson distribution is close to a normal distribution with mean λ and standard deviation $\sqrt{\lambda}$, whose inflection points are at $\lambda \pm \sqrt{\lambda}$; compare $\alpha, \beta \approx \lambda \pm \sqrt{\lambda}$ here. The ratio trick p_r / p_{r-1} is the standard way to find modes of discrete distributions, and it is explored further in Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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